

The Role of Social Identity Labels in CVEs on User Behavior

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ABSTRACT

Psychological, and individual factors like group identity influence social presence in collaborative virtual settings. We investigated the impact of social identity labels, which reflect a user's nation and academic affiliation, on collaborative behavior. In an experiment, $N=18$ dyads played puzzle games while seeing or not seeing such labels. There were no significant differences regarding their social presence, trust, group identification or enjoyment. We argue that social identity labels in dyadic interactions do not change collaborative virtual behavior. We advance the field of sociotechnical applications by highlighting the relationship between psychological characteristics and cooperative behavior in collaborative virtual settings.

Index Terms: Human-centered computing—Collaborative and social computing—Empirical studies in collaborative and social computing; Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality

1 INTRODUCTION

When joining a virtual conference with a crowd of unknown users on social virtual reality (VR) platforms like Mozilla Hubs, users often have to infer info about others. Here, they usually have to rely on the avatar design and a name tag. We hypothesized that knowing more about someone's context, such as their social group membership, would make it easier to navigate such social online events. Social group affiliations refer to social (group) identities and memberships according to Tajfel [8]. Social identities refer to memberships of groups that are considered a "cognitive entity that is meaningful to the subject at a particular point in time" [8].

In offline situations, it has been shown that, for example, more details about a person could make users feel more at ease in various situations and support them in finding social interaction partners based on a "common" ground [7]. Based on that, the topic of this paper is about answering the question *How do social identity labels affect the relationship and cooperative performance among users in collaborative virtual environments?* We explore this with a laboratory study and investigate trust, cooperative user behavior, and social presence in collaborative VR. We identify areas for further study and design suggestions for collaborative VR applications.

2 STUDY DESIGN

We applied a between-subject design for two conditional groups, namely a social group cue labels group and a baseline group. We hypothesize that our social cues lead to improved social presence, trust building and behavior, group identification, task enjoyment, and collaborative performance as opposed to the baseline group.

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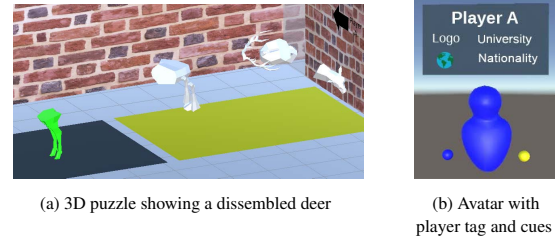


Figure 1: Application design

2.1 Apparatus

For the multi-user VR application, two workstations in two separate laboratory rooms were equipped with an Oculus Rift CV1 and ran our custom multi-user application made with Unity.

2.1.1 Tasks

As a cooperative task, we implemented a three-dimensional puzzle game, inspired by Pan and Steed [6] (see Figure 1a): In the game, each player had a role with unique skills: one player can rotate, and one player can move the puzzle pieces. By that, the roles generate reliance, dependency, and equal contribution. Users had to solve three puzzles of increasing difficulty: They differed in the distinguishability of pieces, the need for manipulation (degrees to rotate, distance to move), and the number of pieces. To evaluate trust, we implemented the Daytrader game [6]: players get 30 tokens each round and decide how many tokens they want to invest as an individual, and how many as a group.

2.1.2 Social Group Cues & Avatars

We adopted information labels that are common in popular social VR apps (see Figure 1b). As social group cues, we chose university affiliation and nationality. Here, we anticipated that the majority of participants would be international students as a result of our recruitment procedures and because previous research had demonstrated that these factors were useful for social identity cues [1, 3]. For the avatars, only a torso and hands are visible to create an abstract body for the avatar (see Figure 1b). We refrained from using accurate models that may encourage prejudice or play to other social identities.

2.2 Experimental Procedure & Measures

After giving informed consent, participants were brought to the labs and prevented from meeting each other before the experiment and were only introduced to each other in VR by highlighting the player tags, the scoreboard, and imaginary out-groups. We stressed that both participants were of the same nationality, the students were from the same university, and they were acting as representatives and performing against another university. Then, they played three rounds of Daytrader before and after the three cooperative puzzles and a discussion.

Pre- and post-experimental questionnaires included questions regarding social presence [5], interpersonal trust ([6], before and after the tasks), group identification [4], and enjoyment [9]. To assess

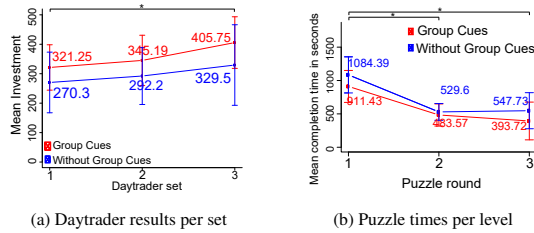


Figure 2: Behavioral measurements

trusting behavior, we considered social dilemma game outcomes [6]. For the collaborative performance, we took the mean completion time of the puzzle game [6]. We report on selected variables of a larger project here.

2.3 Sample

$N = 36$ individuals (61.1% male, 38.9 % female, 0 % diverse) ranging from 19 to 37 years of age ($M = 24.22 \pm 4.08$) participated in the study in a matched sample (convenience sampling via university mailing list and social media). All participants are students with a higher educational background and the majority have never (26.2 %), or rarely (54.8 %) experienced VR before. 50 % of all participants had the German nationality, and 50 % had a different nationality. Eventually, $N = 16$ participants (8 dyads) were assigned to the group with the social group cues, and $N = 20$ to the baseline group (10 dyads).

3 RESULTS

For self-reported trust, puzzle completion and the Daytrader game all between-subject and interaction effects using robust ANOVAs did not show significant differences ($p > .05$).

However, there are few repeated measures significant differences: for the Daytrader results, between the first and third set, and for puzzle completion times between the first and the other levels: The main effect of the sets showed a significant effect ($Q(2, 9.70) = 7.96, p < .01$). Non-parametric Friedman tests showed significant changes ($\chi^2(2) = 8.89, p < .01$) between set 1 and 3 ($r = .55$), but not between set 1 and set 2 or between set 2 and 3 (see Figure 2a). Likewise, for the puzzle completion times only the main effect of the levels showed significant differences ($Q(2, 7.88) = 27.62, p < .001$). Non-parametric Friedman tests showed ($\chi^2(2) = 28.00, p < .001$) that the puzzle completion time changed significantly between levels 1 and 2 ($r = .87$), and level 1 and level 3 ($r = .87$). Both groups completed the first puzzle after about 15 to 20 minutes, the next levels were completed after about 5 to 10 minutes each for both groups (see Figure 2b).

Group identification, social presence, and enjoyment did not differ significantly between groups ($p > .05$) but show all high mean results ($M \Rightarrow 5, \min = 1, \max = 7$).

4 DISCUSSION

We investigated the implementation of social identity cues as labels in a cooperative VR setting and how they impacted the dyadic interpersonal context and the cooperative behavior.

All trust and group identification measures were rather high, independent of condition. One possible explanation could be that users perceived the mere shared purpose for communication in a collaborative virtual environment already as a commonality for a trustful relationship [7]. For puzzle completion times, the results plummet after the first level (see Figure 2b). There were significant differences between the first and other levels which is, however, presumably due to learning effects of game mechanics. Trust benefited from the interaction over time instead of labels. This is similar to findings by

Pan and Steed who observed that trust was always increased after interaction with a team partner [6]. There are some theoretical reasons concerning social identity theory to explain our non-significant results: one requirement for the social identity theory is an internalized membership to a social group for their self-concept [7]. If users are not aware of their groups or do not value them enough, it interrupts crucial psychological processes in social identity theory. Moreover, there are several methodological and ethical limitations in our study: While we carefully designed our study to promote in-group favoritism only and not out-group hostility to avoid harmful effects [7], we showed that social identity labels, although they do not improve the interpersonal relationship, they also do not harm users.

At the same time, we see that despite our matched sample, for future work other social identity cues or more expressive designs (e.g., in avatar design [2, 3]) that lead to possibly more stratification and identification (also for bigger groups and sample sizes) should be considered to investigate the applicability of social identities in VR in collaborative settings.

5 CONCLUSION

Our results suggest that social group cues in labels alone did not lead to differences in user reports or behavior. On the one hand, our results indicate that such labels are no driving forces to leverage trust or social presence in collaborative VR settings. Rather, the time spent together has an impact on the collaboration. On the other hand, we can infer that adding social identity cues in labels also did not harm the relationship between collaborators. For future work we see that crowd investigations are interesting to gain insights in group dynamics [6].

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6 ONLINE RESOURCES

Supplementary material to the experimental procedure can be found at <https://doi.org/10.5281/zenodo.7550695>